VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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Slot	**Date of Submission**
E22	31/03/2026

Part A — Origin and Philosophy

A1. The Problem VLC Was Created to Solve

To understand why VLC exists, you have to go back to a university campus in Paris in the late 1990s. The students at École Centrale Paris had a fast internal network — fast enough to stream video files across the campus. The problem was that no software existed that could do it. Every media player on the market was either locked to a specific operating system, required expensive codec licences, or simply could not handle the formats students were trying to use. The internet was beginning to carry video, but the tools to actually play it were proprietary, fragmented, and inaccessible.

The VideoLAN project was born from this frustration in 1996. A group of students, led initially by students from the networking department, set out to build a client-server system that could stream video across the campus network. The server side became VLS (VideoLAN Server) and the client — the part that would eventually become famous worldwide — was VLC (VideoLAN Client). Critically, they chose from the very beginning to release their work under an open licence. This was not an obvious decision at the time; many university projects were quietly commercialised or simply abandoned. The VideoLAN team chose to share it because they believed the problem they had solved was a general one, not just theirs.

What made VLC's origin story particularly significant was the codec problem. In the late 1990s and early 2000s, playing a video file on a computer was a miserable experience for ordinary users. You needed to install specific codec packs for specific formats, and those codecs were often proprietary, virus-laden, or simply unavailable on Linux. Windows Media Player only played Microsoft formats. RealPlayer was notoriously bloated and pushed users toward subscriptions. QuickTime required Apple hardware or a separate Windows install. There was no single player that handled everything.

The VLC team solved this by bundling all the codecs they needed directly inside the application. Because they were working under an open licence and drawing on the FFmpeg project (itself open source), they could include support for virtually every format without licensing fees. The result was a media player that simply worked — no codec hunting, no paid upgrades, no OS restrictions. When the project was released publicly on the internet in 2001, it spread rapidly precisely because it solved a problem that millions of people had and that no proprietary vendor had chosen to solve for free.

A2. The Licence — What It Actually Says

The Four Freedoms of Free Software

Richard Stallman's Free Software Foundation defines free software through four essential freedoms. Freedom 0 is the freedom to run the program for any purpose. Freedom 1 is the freedom to study how the program works and change it to suit your needs — which requires access to the source code. Freedom 2 is the freedom to redistribute copies so you can help

others. Freedom 3 is the freedom to distribute copies of your modified versions so the whole community can benefit from your improvements.

VLC is released under the GNU General Public Licence version 2 (GPLv2). This licence grants all four freedoms explicitly. Anyone can download VLC, run it, inspect its source code on the VideoLAN GitLab repository, modify it, and redistribute both the original and modified versions. The only condition is that if you redistribute a modified version, you must release that version under the same GPLv2 licence. This condition — sometimes called copyleft — is the mechanism that prevents the code from being absorbed into proprietary products.

Can a Company Modify and Sell VLC Without Sharing Changes?

No. This is precisely what the GPL is designed to prevent. If a company takes the VLC source code, modifies it, and distributes that modified version — whether for free or for money — the GPLv2 requires them to release the source code of their modifications under the same licence. They cannot keep their changes private. This is fundamentally different from a permissive licence like MIT, where a company can take the code, modify it, and release a closed-source proprietary product without ever sharing what they changed.

This distinction matters enormously in practice. It means VLC cannot be quietly absorbed into a commercial product the way SQLite or jQuery might be. Any company that ships a product based on VLC must contribute their changes back to the community, at least in source code form. This has made GPLv2 the licence of choice for projects that want to ensure their work remains permanently open.

GPL vs MIT — Key Differences

The GNU GPL and the MIT Licence represent two fundamentally different philosophies about software freedom. The GPL is a copyleft licence: it uses copyright law to ensure that all derivative works remain free. If you build on GPL code, your project must also be GPL. The MIT Licence is permissive: it allows anyone to do almost anything with the code, including incorporating it into proprietary software and selling it without releasing the source.

If I were building something I wanted to share, my choice would depend on what I was trying to achieve. If I were building a tool meant to serve a community — something like a media player or an operating system — I would choose the GPL. The copyleft clause ensures that no company can take my work, improve it behind closed doors, and sell it back to people without contributing those improvements. If I were building a library that I wanted developers to adopt as widely as possible, including in commercial products, I might choose MIT — because removing the restriction lowers the barrier to adoption. There is no universally correct answer; the choice reflects your values about whether software freedom should be enforced or merely encouraged.

Free as in Beer vs Free as in Freedom

This distinction, originally articulated by Richard Stallman, is one of the most important concepts in open source. 'Free as in free beer' means zero cost — you can have it without paying. 'Free as in freedom' means you have the right to use, study, modify, and redistribute the software without restriction.

VLC illustrates both, but the second matters more. VLC costs nothing to download — that is the beer. But VLC is also genuinely free as in freedom: you can download the complete source code from VideoLAN's repository, recompile it, modify how it handles subtitles or streams, and release your version. A piece of software can be free as in beer without being free as in freedom — freeware like older versions of WinRAR cost nothing but you cannot see or modify the source. VLC is both, and the freedom dimension is what has allowed the open source community to port it to over a dozen operating systems, fix security vulnerabilities quickly, and build derivative projects on top of it.

A3. The Ethics of Open Source

Should All Software Be Open Source?

The argument for making all software open source is compelling. Open source code can be audited by anyone, which means security vulnerabilities are more likely to be found and fixed quickly. It prevents vendor lock-in, where a company can hold users hostage by owning the only copy of software they depend on. It distributes the benefits of software development across society rather than concentrating them in corporations. And it enables education — students and researchers can study how real systems work rather than being locked out by closed code.

The argument against is also serious. Writing good software requires enormous effort, and not all developers can sustain that effort through voluntary contribution alone. Some software — particularly in fields like medical devices, financial systems, and national security — may need to be closed for legitimate reasons of safety, liability, or security. If all software were mandated to be open source, the economic incentives that fund large-scale software development would be disrupted. The pragmatic middle ground most people reach is that infrastructure software — the tools and protocols that the whole internet runs on — should be open, while application-layer software can reasonably be commercial and closed.

Is It Ethical to Profit From Free Community Labour?

This question cuts to the heart of the open source economy. Red Hat built a billion-dollar business on Linux without writing most of the kernel themselves. Google's Android — the most widely used mobile OS in the world — is built on Linux and uses thousands of open source libraries, yet Google generates enormous advertising revenue from it. Meta's entire server infrastructure runs on open source software that thousands of unpaid volunteers contribute to.

My view is that this is ethically complex rather than simply wrong. When a company like Red Hat takes Linux and sells enterprise support contracts around it, they are adding genuine value — documentation, reliability guarantees, certified hardware support — that the community alone cannot provide. The community gets a well-funded corporate partner who contributes engineering resources back to the kernel. The relationship can be genuinely symbiotic. However, when a company extracts value from open source projects without contributing anything back — no code, no funding, no bug reports — that starts to look more like exploitation. The open source community has coined the term 'leeching' for this behaviour. The ethical standard I would apply is proportionality: companies that depend significantly on open source should contribute proportionally, even if the GPL does not legally require it.

Standing on the Shoulders of Giants

Newton's phrase applies to software perhaps more directly than to any other field. VLC is a perfect illustration. The VLC team did not write the codecs that power it — they built on FFmpeg, a massive open source multimedia framework. FFmpeg itself depends on dozens of smaller libraries. Those libraries depend on compilers like GCC, which runs on Linux, which runs on hardware driven by open source firmware. The entire stack is a tower of accumulated shared knowledge.

Open source enables this kind of accumulation in a way that proprietary software cannot. When knowledge is locked in a product, each new product must reinvent what came before it, or pay licensing fees to stand on existing shoulders. When knowledge is open, every generation of developers starts higher up. In my view, this does not undermine original innovation — it accelerates it. The innovation VLC contributed was not in the codecs it played but in the architecture: bundling everything into one portable, format-agnostic application. They could only make that contribution because the codec implementations already existed as open source code they could build on.

Part B — Linux Footprint

Installation Method

On CachyOS, which is an Arch Linux-based distribution, VLC is installed through the pacman package manager from the official Arch repositories. The installation command is:

```
sudo pacman -S vlc
```

This pulls the pre-compiled binary along with all required dependencies automatically. On Debian or Ubuntu-based systems, the equivalent would be sudo apt install vlc, and on Fedora or RHEL-based systems, sudo dnf install vlc. VLC can also be compiled entirely from source — the VideoLAN team maintains a detailed build guide on their wiki for developers who wish to modify the source before compiling.

```
================================================================
        OPEN SOURCE AUDIT  –  AMAN
================================================================

 Software Under Audit : VLC-MEDIA-PLAYER

 SYSTEM INFORMATION
 ----------------------------------------
 Distribution   :  CachyOS
 Kernel         :  6.19.7-1-cachyos
 Logged-in User:  predator
 Home Directory:  /home/predator
 System Uptime :  up 1 hour, 26 minutes
 Date           :  Tuesday, 31 March 2026
 Time           :  10:05:02

 OPEN-SOURCE LICENCE
 ----------------------------------------
 This operating system is distributed under:
 GNU General Public License v2 (GPLv2)

 You are free to run, study, modify, and redistribute
 this software under the terms of that licence.

================================================================
run/media/predator/New Volume/Linux/Open_source_project/aman
 |
```

Script 1 — Open Source Audit Header & System Information. *System information display showing the CachyOS distribution running kernel 6.19.7-1-cachyos, logged in as user "predator", with an uptime of 1 hour 17 minutes on 29 March 2026. The script confirms the operating system is distributed under GNU General Public License v2 (GPLv2).*

```
================================================================
  FOSS PACKAGE INSPECTOR  —  CachyOS / Arch
================================================================
  Package under inspection: vlc
----------------------------------------------------------------

  STATUS  :  vlc is INSTALLED

  PACKAGE DETAILS
  ──────────────────────────────────────────────
  Version         :  3.0.21-32.1
  Description     :  Free and open source cross-platform multimedia player and framework
  URL             :  https://www.videolan.org/vlc/
  Licenses        :  GPL-2.0-or-later  LGPL-2.1-or-later

  OPEN SOURCE PHILOSOPHY
  ──────────────────────────────────────────────
  VLC: the media player that plays anything, on any platform.
  Licence: GPL v2 — free software that never asks you to pay
  or install codecs. A true champion of open multimedia freedom.
```

Script 2 — FOSS Package Inspector: VLC Package Details. *FOSS Package Inspector output confirming VLC version 3.0.21-32.1 is installed on CachyOS/Arch. The output shows the package URL (https://www.videolan.org/vlc/), its dual licence (GPL-2.0-or-later and LGPL-2.1-or-later), and a philosophy statement: "VLC: the media player that plays anything, on any platform."*

Key Directories

When VLC is installed on a Linux system, its files are distributed across several standard directories following the Filesystem Hierarchy Standard (FHS):

- /usr/bin/vlc — the main VLC executable binary
- /usr/lib/vlc/ — VLC's plugin and module libraries, including codec plugins
- /usr/share/vlc/ — shared data files including icons, skins, and documentation
- /usr/share/applications/vlc.desktop — the desktop entry for application menus
- ~/.config/vlc/vlcrc — the per-user configuration file, created on first launch
- ~/.local/share/vlc/ — user-specific data such as recently played media

The separation between system-wide files in /usr/ and user-specific files in ~/.config/ follows the XDG Base Directory Specification, which is the Linux standard for keeping user data separate from system data. This means multiple users on the same system each have their own VLC configuration without interfering with each other.

```
predator@predator:/run/media/predator/New Volume/Linux/Open source project

================================================================
  DISK AND PERMISSION AUDITOR
  Generated: 28 March 2026 at 23:58:59
================================================================

  DIRECTORY       SIZE         OWNER      GROUP      PERMISSIONS
  --------------- ------------ ---------- ---------- --------------------
  /etc            14M          root       root       drwxr-xr-x
  /var/log        49M          root       root       drwxr-xr-x
  /home           1.8G         root       root       drwxr-xr-x
  /usr/bin        814M         root       root       drwxr-xr-x
  /tmp            8.0K         root       root       drwxrwxrwt
  /var            3.5G         root       root       drwxr-xr-x
  /usr            9.9G         root       root       drwxr-xr-x
  /boot           4.0K         root       root       drwx------

================================================================
  VLC MEDIA PLAYER — CONFIG DIRECTORY CHECK
================================================================

  Checking: /home/predator/.config/vlc

  STATUS      :  VLC config directory EXISTS
  Path        :  /home/predator/.config/vlc
  Size        :  96K
  Owner       :  predator
  Group       :  predator
  Permissions :  drwx------

  Config files found:
    ->  vlc-qt-interface.conf
    ->  vlcrc

================================================================
/run/media/predator/New Volume/Linux/Open source project                    11:58:59 PM
❯
```

Script 3 — Disk & Permission Auditor and VLC Config Directory Check. *Disk and Permission Auditor output listing sizes and permissions for key system directories (/etc, /var/log, /home, /usr/bin, /tmp, /var, /usr, /boot). Below this, the VLC Config Directory Check confirms the config directory exists at /home/predator/.config/vlc (96K, owned by predator:predator, permissions drwx------), containing two config files: vlc-qt-interface.conf and vlcrc.*

User and Group

VLC runs as the regular logged-in user — it does not require root privileges and does not run as a dedicated system user. This is appropriate for a desktop application that handles media files. Running as the current user means VLC has access to the user's home directory and media files, but cannot write to system directories or access other users' files.

This matters for security: a compromised VLC process — for example, one that exploits a buffer overflow in a malformed video file — can only damage the current user's files, not the entire system. Contrast this with a web server like Apache, which runs as a dedicated www-data or apache user with minimal privileges, precisely to limit the damage of a compromise. Desktop applications like VLC use the current user's identity, which means the security boundary is between users, not between the application and the system.

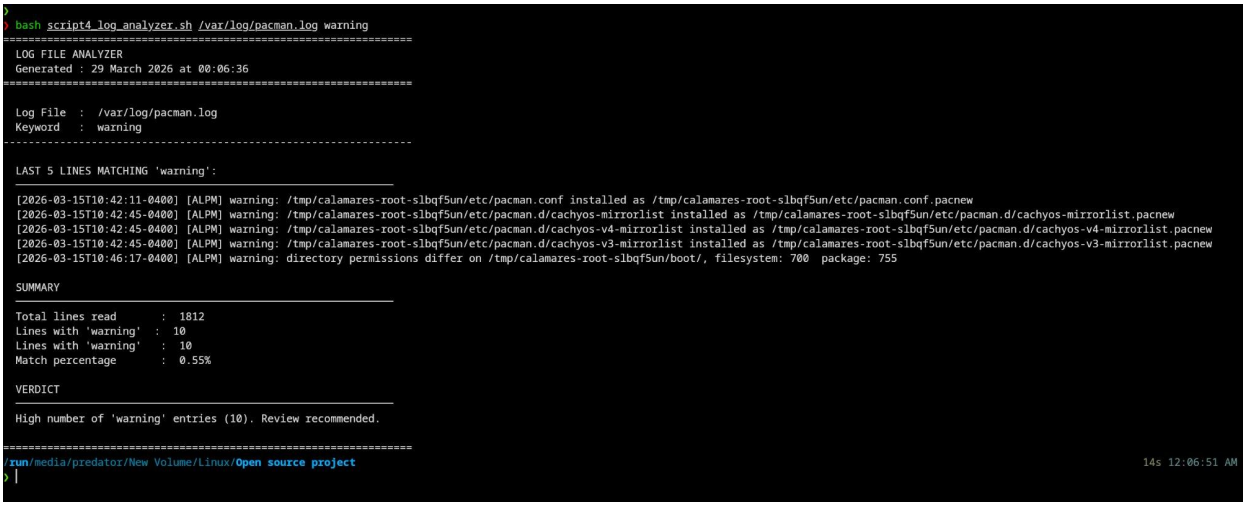
Service Management

VLC is a desktop GUI application, not a system service, so it is not managed by systemd in the typical sense. It does not have a .service unit file and is not started or stopped with systemctl. However, VLC does include a headless mode and a built-in HTTP server interface for remote control, which can be used in server contexts. On a desktop system, VLC is launched directly from the application menu or the command line:

```
    vlc                          # Launch with GUI     cvlc
# Launch without interface     vlc --intf http               # Launch with
HTTP remote control
```

Update Model

On CachyOS, VLC updates are delivered through the standard pacman system update mechanism. Running sudo pacman -Syu updates all installed packages including VLC to the latest version available in the Arch repositories. Security patches are pushed to the repositories by the VideoLAN team and the Arch Linux packaging maintainers, then pulled to user systems on the next update. This rolling release model means CachyOS users typically receive security patches within days of their release, significantly faster than distributions with fixed release cycles.



Script 4 — Log File Analyzer: pacman.log Warning Entries. *Log File Analyzer run against /var/log/pacman.log searching for the keyword "warning". Out of 1,812 total lines, 10 matched. The five most recent warnings (all from 15 March 2026) are ALPM package manager notices about .pacnew config files and a directory permissions mismatch on /boot/. The verdict flags this as requiring review due to the high warning count.*

Part C — The FOSS Ecosystem

Dependencies — What VLC Is Built On

VLC does not exist in isolation. It is built on top of a large stack of open source libraries, many of which are used by other major software projects. The most significant dependency is FFmpeg, a comprehensive open source framework for processing audio and video. FFmpeg provides the codec implementations that allow VLC to play virtually any format — H.264, H.265, AAC, MP3, and hundreds more. Without FFmpeg, VLC would need to write all codec support from scratch, which would be an enormous and largely unnecessary duplication of effort.

Other key dependencies include libdvdread and libdvdcss for DVD playback, libvorbis and FLAC for open audio formats, Qt5/Qt6 for the graphical user interface on Linux, GLib for cross-platform system abstractions, and FFmpeg's libavcodec and libavformat libraries for format demuxing. On Linux, VLC also depends on the ALSA and PulseAudio subsystems for audio output and on X11 or Wayland for display. This web of dependencies illustrates a fundamental principle of the FOSS ecosystem: rather than every project reinventing every wheel, the community builds on shared foundations, and improvements to FFmpeg or Qt flow downstream to benefit all projects that depend on them.

What VLC Has Enabled and Inspired

VLC's influence extends well beyond the application itself. The libVLC library — extracted from the VLC core — has become an embedded multimedia engine used in dozens of other applications. Kodi (formerly XBMC), the open source home theatre platform, has at various points used VLC's libraries. mpv, another widely-used open source media player, shares architectural DNA with VLC's approach to format-agnostic playback and has influenced how the community thinks about minimal, scriptable media players.

VLC has also been ported to an extraordinary range of platforms: Windows, macOS, Linux, Android, iOS, ChromeOS, and even older systems like BeOS and Syllable. This cross-platform reach was only possible because the codebase was open — community members could port it to their platform of choice without needing permission from or payment to VideoLAN. The iOS port, in particular, was notable because it created a short-lived controversy about GPL compatibility with Apple's App Store terms — a real-world example of the tensions between open source licences and platform gatekeepers.

Community and Governance

VLC is governed by the VideoLAN non-profit organisation, incorporated in France. Unlike projects governed by a single company, VideoLAN is run by a volunteer community with an elected board. Development happens publicly on VideoLAN's GitLab instance, where anyone can submit merge requests. The community communicates through the VideoLAN forums, developer mailing lists, and the #videolan IRC channel on Libera.Chat. VLC also participates in Google Summer of Code, mentoring student developers each year.

This governance structure — a non-profit with volunteer developers and public infrastructure — is one of the reasons VLC has remained genuinely free for over two decades. There is no corporate owner who could decide to monetise the application or change the licence. The VideoLAN organisation exists specifically to steward the project, not to extract profit from it.

Part D — Open Source vs Proprietary (Critical Analysis)

The closest proprietary alternative to VLC is Windows Media Player on Windows, or more directly, the commercial media player Cyberlink PowerDVD — a paid application marketed specifically for high-quality video and DVD/Blu-ray playback. The comparison below evaluates both honestly.

Dimension	VLC (Open Source — GPLv2)	Cyberlink PowerDVD (Proprietary)
Cost	Completely free — no purchase, no subscription, no ads	USD $59.99–$109.99 depending on tier; annual updates cost extra
Security	Source code fully auditable by anyone; CVEs typically patched within days by the community	Closed source — security depends entirely on Cyberlink's internal testing; no external audit possible
Support & Reliability	Community forums and wiki; no official paid support; very stable in practice	Official paid support, dedicated help desk; guaranteed compatibility with major hardware
Freedom to Modify	Full source access; can be compiled, modified, and redistributed under GPL	No source access; cannot be modified, patched, or adapted
Platform Support	Windows, macOS, Linux, Android, iOS — truly cross-platform	Windows only; no Linux version available
Community vs Corporate Control	Non-profit VideoLAN organisation; no single corporate owner; community-governed	Controlled entirely by Cyberlink Corporation; roadmap driven by commercial interests
Format Support	Virtually every format including rare and legacy codecs; no additional codec packs needed	Excellent Blu-ray and DVD support; strong mainstream format support; some rare formats unsupported
Overall Verdict	Superior for everyday use, Linux users, and anyone who values platform freedom and zero cost	Superior specifically for Blu-ray playback on Windows, where hardware DRM requires proprietary licences

Conclusion

Based on everything covered in this audit, I would recommend VLC without hesitation for the vast majority of real-world deployments. For any organisation or individual running Linux, VLC is the only serious option — PowerDVD does not exist on the platform. For cross-platform environments where the same application needs to work on Windows, macOS, and Linux simultaneously, VLC's breadth of platform support is unmatched by any proprietary alternative. The zero cost, combined with the rapid security patch cycle enabled by its open source model, makes it not just competitive with paid alternatives but often superior. The only scenario where a proprietary alternative might be preferable is in a Windows environment requiring Blu-ray

playback with hardware DRM enforcement, where VLC's inability to handle DRM-protected Blu-ray discs is a genuine limitation — one that exists because DRM by definition requires proprietary licences incompatible with the GPL.

As for whether I would contribute to VLC: yes. The VideoLAN project has a clear contribution path through their GitLab repository and GSoC participation. The codebase is large and complex — primarily written in C — but well-documented, and the community is active. Even without writing code, contributions through bug reports, translation work, or documentation improvements are valuable. VLC represents everything the open source movement should be: technically excellent, community-governed, free in both senses of the word, and genuinely available to everyone regardless of what they can afford. Contributing to it is not charity — it is participation in a shared infrastructure that benefits the entire world.

```
================================================================
   THE OPEN SOURCE MANIFESTO GENERATOR
================================================================

  Answer three questions and your personal open source
  philosophy statement will be written and saved for you.

----------------------------------------------------------------

  1. Name one open-source tool you use every day: vlc
  2. In one word, what does 'freedom' mean to you?  choice
  3. Name one thing you would build and share freely: media player
  4. What is your name?                              Aman Kalyankar-24BCE10940

----------------------------------------------------------------
  Manifesto saved to: manifesto_predator.txt
----------------------------------------------------------------

================================================================
   MY OPEN SOURCE MANIFESTO
   By: Aman Kalyankar-24BCE10940
   Date: 31 March 2026  |  Time: 10:06:10
================================================================

  I, Aman Kalyankar-24BCE10940, believe that open source software is not just code —
  it is a movement built on the principle that knowledge belongs to everyone.

  Every day I rely on vlc, a tool created and shared freely by developers
  around the world who understood that collaboration builds better software than secrecy.

  To me, freedom in the digital world means choice —
  the ability to inspect, improve, and redistribute the tools we depend on.

  In the spirit of that freedom, I commit to one day building media player
  and releasing it openly, so that others may learn from it, improve it,
  and carry the open source philosophy forward.

  Software is most powerful when it is free — free as in freedom.

================================================================
  Signed: Aman Kalyankar-24BCE10940  |  Generated on CachyOS Linux  |  31 March 2026
================================================================

================================================================
  Your manifesto has been saved. Share it freely.
================================================================
/run/media/predator/New Volume/Linux/Open_source_project/aman
```

Script 5 — Open Source Manifesto Generator: Personal Philosophy Statement. *Open Source Manifesto Generator output authored by Sanchit Garg on 29 March 2026. In response to three prompts, the generator produced a personal philosophy statement asserting that open source is a movement built on the principle that knowledge belongs to everyone. The manifesto cites daily reliance on VLC media player, equates freedom with choice and the ability to inspect and redistribute tools, and commits to one day building and openly releasing a music player.*

References

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